

Sailing by Paintapu's field

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Abstract

Every floating-point number carries a coarse logarithm for free: read its bit pattern as an integer and, up to scale, you get $E + m$. The true logarithm differs by a small residual $\varepsilon(m) = \log_2(1 + m) - m$, the same bump on every binade. This paper asks whether ε can be exactly corrected by any finite structural means, and shows the natural structural routes close. Three results anchor the argument: on the binary hyperbolic tiling, no corrector of bounded corona radius matches ε on every cell; the level-by-level Padé construction on the Stern–Brocot scaffold admits no fixed finite-dimensional linear update; and ε is transcendental at every interior dyadic. Around these, polynomial approximation, affine clock changes, and invariant-measure aggregation each fail for its own reason.

“On two occasions I have been asked, — ‘Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?’ . . . I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question.”

— Charles Babbage

“You best start believing in ghost stories, Miss Turner . . . you’re in one.”

— Hector Barbossa

Scientific notation is introduced early on in education. It is a load-bearing representation in any sense of architecture which bears a load. Written in scientific notation, Avogadro's number and the Planck length are equally compact despite being fifty orders of magnitude apart. In nearly all cases we illustrate three facts about this notation in the following order:

1. *Scale is easy*. The two numbers 6.02×10^{23} and 1.616×10^{-35} fit on the same line.
2. *Multiplication is easy*. Checking “about fifty orders of magnitude” is $23 + 35$.
3. *Addition is hard*. Adding Avogadro's number to the Planck length is a mess. Nobody says why.

Students who ask why this must be are given a hoary allegation that the situation is “opposite” in a positional number system. Those who question such an assertion are told this is just how it is. For many, this is first contact with the beautiful inquisitiveness of the Sciences. We ask, if that is just how it is, why must it be *just* like that? Why must there be an irreducible cost to addition in this logarithmic system?

Within the realm of the Reals, I cannot tell you the answer now; the voyage changes you into someone who cannot be told otherwise. In the Surreal numbers, however, the answer is simple. It *must* be just like that; there must be an irreducible cost to addition because for all interior $m = k/2^p$,

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$0 < k < 2^p$, the surreal birthday of $\varepsilon(m)$ (where $\varepsilon(m) = \log_2(1 + m) - m$) is ω . No finite birthday suffices. We proceed from statement here to proof alongside similar proofs and yarns in the Reals.

Good sea stories and good proofs share one quality: you don't need to believe for them to be true, but it helps. If you have been lost in a tree and found your way back, you know that horizontal movement is expensive. If you have followed one irrationality proof closely enough to remember the moment it snapped shut, you know what “never” means. If you have felt two clocks disagree — music against a metronome, jet lag, any moment where time is out of phase — you know lineal correction is not as easy as it seems.

§0. The poor man's logarithm

Scientific notation in base 10 writes $x = a \times 10^E$ with $1 \leq a < 10$, but a base is a choice. In base 2, we may write $x = (1 + m) \times 2^E$ with $m \in [0, 1)$. The exponent E is an integer. The mantissa m is a binary fraction. Everything that follows takes place in base 2.

A floating-point number x in the binade $[2^E, 2^{E+1})$ is stored as an exponent E and a mantissa $m \in [0, 1)$, so that $x = 2^E \cdot (1 + m)$. The bit pattern encodes E and m separately. If we read the entire bit pattern as an integer, up to bias and scaling the result is $E + m$. Mitchell (1962) observed that $E + m$ is a coarse logarithm of x . The representation contains its own log table.

Velvel Kahan exploited this. To compute a square root, take the logarithm, divide by two, take the antilogarithm. With a slide rule this requires three table lookups. In base-2 floating point the first and third are free — reinterpret the bit pattern as an integer, operate, reinterpret back — and the second is a bit shift. Kahan's “magic square root” is the slide rule method executed on the representation itself, with no table in memory. The representation is the table.

This is not an approximation that happens to work. Day (2023) proves that $L(x) = E + m$ is the unique affine function on the binade that agrees with $\log_2(x)$ at both endpoints. It is the optimal coarse surrogate in a precise sense: the endpoint-exact affine map that minimizes the worst-case deviation from $\log_2$ on $[2^E, 2^{E+1})$.

The deviation has a name. Define

$$\varepsilon(m) = \log_2(1 + m) - m, \quad m \in [0, 1). \quad (1)$$

Then $\log_2(x) = L(x) + \varepsilon(m_x)$. The function ε is zero at $m = 0$ and $m \rightarrow 1$, positive on $(0, 1)$, concave, with a unique maximum at $m^* = 1/\ln 2 - 1 \approx 0.4427$, where $\varepsilon(m^*) \approx 0.0861$. It is determined entirely by $\ln 2$.

```
i  = 0x5f3759df - ( i >> 1 );           // inarticulate sailor cursing
```

The constant `0x5f3759df` computes $x^{-1/2}$ by the method of log tables: reinterpret the bit pattern as an integer (the log is free), shift right and subtract (divide by -2 in log-space), reinterpret back (antilog). One Newton iteration refines. The constant encodes the exponent bias (127), the mantissa width (23), and one real parameter: the optimal intercept σ , a functional of ε . Its value depends on the shape of the bump — its endpoints, its maximum, its concavity.

One number suffices because the bump is the same on every binade. The binary subdivision of $[0, 1]$ to depth d creates 2^d cells. In the Poincaré half-plane, depth d lives at height $y = 2^{-d}$. Each cell has Euclidean width $1/2^d$ and horocyclic arc length 1, independent of d . Consecutive depths are separated by hyperbolic distance $\ln 2$. Every cell is congruent: horocyclic width 1, geodesic height $\ln 2$. The function ε lives on this cell and does not change from binade to binade: same shape, same peak, same zeros. The optimal intercept is the same on every cell because there is only one cell shape.

Day's decoupling theorem: the coarse stage (computing L) and the correction stage (approximating the residual) are independent problems. The coarse solution is inherited by every correction architecture. The correction stage approximates $z^{-1/b}$ on an interval $[z_{\min}, z_{\max}]$, where the extrema of z are values of $2^{\varepsilon(r/k)}$ at Farey sample points — the mediants of the Stern–Brocot tree. Day's extremal classification is optimization in ε -coordinates. The Stern–Brocot subdivision and the binary mantissa subdivision are two different metrics on $[0, 1]$; the level-by-level discrepancy between them is §4's object.

ε is our object. Everything after this is about what happens when you try to make it vanish by finite structural means.

§ 1. Polynomial packing on a tile

The congruence of the cell — the prototile — lets a single integer absorb the bounds of ε . The coarse stage is free. The natural next move is to improve on it: approximate the correction $z^{-1/b}$ with a polynomial, driving the error toward zero. Each correction is a local computation — optimal within its tile, at its precision. But even this local optimality is fragile: the same packing tiled two ways can yield different density values (Bowen 2002, §1.2.4).

Grant the machine an unbounded real-valued accumulator. It reads the bit string $b_1, b_2, \dots, b_d$ with positional weights $\delta_j = 2^{-j}$ and computes $L(x) = E + m_x$ exactly. No finite-configuration aliasing argument applies: the accumulator can produce 2^d distinct outputs despite having one control state.

The correction stage remains. Having computed $L(x)$ exactly, form the coarse-stage variable $z(x) = x^a \cdot y(x)^b$, where y is the coarse approximation to $x^{-a/b}$. Day shows that z is a bounded continuous periodic function of $L(x)$ with period b , and its range is a closed interval $[z_{\min}, z_{\max}]$. Write $\rho = z_{\max}/z_{\min}$. The correction seeks to approximate $z^{-1/b}$ on this interval. If one restricts to a polynomial corrector $p(z)$ of degree at most n , the minimax relative error is

$$\varepsilon_n^*(\rho) = \min_{\deg p \leq n} \max_{z \in [z_{\min}, z_{\max}]} \frac{|z^{-1/b} - p(z)|}{z^{-1/b}}. \quad (2)$$

For every finite n and every nondegenerate interval, this error is strictly positive. The function $z^{-1/b}$ is not a polynomial on any interval of positive length. The polynomial wall stands: $\varepsilon_n^*(\rho) > 0$ for every finite n . This pillar survives unbounded memory; §3 will give the local counterpart in corona form.

Surreal §1. A surreal number x born on day d has sign expansion $s: d \rightarrow \{+, -\}$. Day $d < \omega \implies x$ is a dyadic rational. Day $\omega \implies$ sign expansion is infinite.

§ 2. Affine recurrence

Polynomial correction defeated a static approach: fix a degree, approximate, accept a nonzero residual. The basis was not rich enough no matter the degree. A different strategy: instead of adding polynomial coefficients, track the Fourier modes of ε directly and average the correction into convergence.

The Baumslag–Solitar group $BS(1, 2)$ has two generators. The generator a acts by multiplication: $x \mapsto 2x$. The generator b acts by addition: $x \mapsto x + 1$. The Cayley graph of $BS(1, 2)$ is the binary tiling. Every computation that combines additions and multiplications traces a path on this graph. Multiplication moves along geodesics. Addition moves along horocycles. The machine adds numbers. Each addition is one tick of additive time.

Under multiplication, mantissa distributions converge exponentially to the logarithmic distribution. The geodesic direction and the mantissa coordinate are matched: multiplication is the natural operation for mantissas. Under addition, they do not converge (Schatte, Lemma 1). At each step, the mantissa of the running sum rotates — the phase of the distribution shifts by approximately $1/n$ at step n , since $\log(n+1) - \log(n) \approx 1/n$. Early steps rotate fast. Late steps rotate slowly. The rotation never stops.

Try to average the rotation away. Define k -fold Cesàro means: $\mathfrak{A}_0(a_n) = a_n$ and $\mathfrak{A}_{k+1}(a_n) = (1/n) \sum_{m=1}^n \mathfrak{A}_k(a_m)$. By Schatte's Lemma 6, the leading term of $\mathfrak{A}_k(h_n(r))$ at frequency r has modulus $|2\pi ir + 1|^{-k}$ and argument $2\pi r \log(|a|n) + k \cdot \arg(2\pi ir + 1)^{-1}$. The modulus shrinks with k . The argument still rotates. The phase $2\pi r \log(|a|n)$ increases without bound. For every k , there exist arbitrarily large n_1, n_2 with phases separated by at least $\pi/2$. The separation in sup norm is at least $c \cdot |2\pi i + 1|^{-k}$ for an absolute constant $c > 0$. The k -th Cesàro means are not Cauchy.

Riesz logarithmic means converge (Schatte 1986, Theorem 5). Define $\mathfrak{L}(a_n) = (1/\log n) \sum_{j=1}^n a_j/j$. Schatte proves $|\mathfrak{L}(M_n(x)) - \log x| \leq A_9(\sigma^2/a^2 + 1)/\sqrt[3]{\log n}$, uniformly in $1 \leq x \leq e$. The bound vanishes as $n \rightarrow \infty$. The Riesz method converges with no residual.

The operative difference is the weighting. Cesàro weights each term uniformly by $1/n$. Riesz weights the j -th term by $1/j$. The weight $1/j$ matches the deceleration of the rotation: at step j , the phase increment is approximately $1/j$, so the weight gives each step influence proportional to how much new phase information it carries. The Riesz weight $1/j$ is the Jacobian of the coordinate change from additive time to multiplicative time. In multiplicative time — $\log 1, \log 2, \log 3, \dots$ — the rotation is approximately uniform. Averaging uniformly in log-time works, and averaging uniformly in log-time is averaging with weight $1/j$ in linear time.

To cross from Cesàro to Riesz, the machine must change its weighting from uniform to logarithmic. The machine ticks in additive time. The mantissa lives in multiplicative time. The coordinate map between the two clocks is $\psi(m) = \log_2(1+m)$. Its deviation from the identity clock is ε .

The machine's operations on the mantissa are affine within a binade: additions translate by a constant, multiplications scale by a constant. (Cross-binade carries renormalize and are set aside as a coarse-grain model.) These generate the group $\text{Aff}^+(\mathbb{R})$ of maps $x \mapsto ax + b$ with $a > 0$. The coordinate change $\psi(m) = \log_2(1+m)$ is not affine — it is logarithmic. The composition of affine maps is affine: $\text{Aff}^+(\mathbb{R})$ is a group. No finite composition of the machine's operations produces ψ . §4 runs this same closure-under-composition argument one level of generality up, in a finite-dimensional matrix algebra.

Surreal §2. Walker clock: $t \in \mathbb{N}$, one bit per tick. Tree clock: $\tau = \log_2(1+m)$, $m \in [0, 1)$. Drift: $\varepsilon(m) = \tau - m$.

§ 3. Corona crawl

Polynomial correction (§1) and affine recurrence (§2) were both defeated by the structure of the problem — one by the spectral shape of ε , the other by the clock mismatch between additive and multiplicative time. Both used the binary tiling as background. Now we use it as the instrument.

The congruent cell — horocyclic width 1, geodesic height $\ln 2$ — embeds in the Poincaré half-plane as a tile. This is Bowen's stripe model (2002, §1.2.2) with strip width $W = \ln 2$. The binary subdivision of $[0, 1]$ to depth d produces 2^d such tiles corresponding to the standard dyadic subdivision. These tilings are not directionless. Each one points toward the ideal boundary by a continuous, equivariant protrusion direction (Bowen 2002, §2.3.1). Any attempt to average cells as though they were freely interchangeable will have to answer to that asymmetry.

Each cell at depth d carries a Dolbilin–Frettlöh tail (2010, Definition 3.4) — a horocyclic-local

sign sequence in $\{-1, +1\}^{\mathbb{N}}$ encoding the cell's position within its ring, distinct from the dyadic descent address. Its length- k prefix (the k -tail) determines the cell's k -corona combinatorial type. By Dolbilin–Frettlöh's Proposition 4.3, the binary tiling has $N_k = 2^{k-1}$ congruence classes of k -coronae. The tiling is non-crystallographic (Theorem 4.4).

Identify each depth- d cell with its left endpoint $k/2^d$ as canonical mantissa. A k -local corona-invariant corrector is a map on cells whose value depends only on the k -corona congruence class; it is exact if it equals ε at every cell's canonical mantissa. This model is strictly weaker than a bounded-state streaming machine — the streaming machine reads the dyadic address bit by bit, while the corona model sees only a finite congruence invariant of the local tiling.

No k -local corona-invariant corrector on the depth- d cells is exact once $d \geq k + 1$. For $k = 1$, $N_1 = 1$: every cell sits in one corona class, so the corrector outputs a single value c . The cells at depth $d \geq 2$ include both 0 and $1/2$ as left endpoints, with $\varepsilon(0) = 0 \neq \varepsilon(1/2) = \log_2(3/2) - 1/2 \approx 0.085$, so no constant matches both.

For $k \geq 2$, the orientation-reversing isometry of Prop 4.3 pairs each k -tail $(s_1, \dots, s_k) \in \{\pm 1\}^k$ with its bit-complement; the involution is fixed-point-free ($s_j \in \{\pm 1\}$ forbids $s_j = -s_j$), so the 2^k tail patterns split into 2^{k-1} isometry orbits. Tail patterns cycle along the depth- d ring with period 2^k , so each orbit contributes 2^{d-k+1} cells — at least 4 for $d \geq k + 1$. At most one cell per class has left endpoint 0, leaving at least three with left endpoints in $(0, 1)$. Strict concavity of ε on $(0, 1)$ makes every horizontal line meet the graph in at most two points; among three distinct interior mantissas at least two distinct ε -values appear, and a single output value cannot match all three.

Surreal §3. $N_k = 2^{k-1}$ corona classes (Dolbilin–Frettlöh). 2^{d-k+1} cells per class via orientation pairing. ε strictly concave on $(0, 1)$. Local aliasing at $d \geq k + 1$. Walker width q : open.

§ 4. The Padé ghost

Polynomial correctors failed at every finite degree (§1). We have only a faint hope. We can call on the ghost of the greatest dead numerical analyst Henri Padé. If we get him close enough, maybe his genuinely unearthly approximants can go the distance. Padé approximants are rational functions — ratios of polynomials — fitted to match as many terms of a power series as their degrees allow, reaching past the radius where polynomial truncation fails.

The bijection between the two subdivisions is Minkowski's $?(x)$, continuous and increasing, each finite level of its construction absolutely continuous, the limit singular (Salem 1943). If the corrections between levels could be made to close, the singular limit would be reachable from finite data — Padé's best chance.

Build $?(x)$ level by level from the Stern–Brocot tree. At each level, deform the metric on each Farey interval to match the corresponding dyadic interval. The level- n floor is absolutely continuous by construction. Integration against it works. On this scaffold, finite closure would be straightforward: the grids re-align after finitely many levels, and the corrections repeat. Rational branching ratios on a tree whose geometry has been aligned at every finite level gives Padé the best possible perch.

Let C_k be the level- k floor of the construction above — the absolutely continuous interpolant of $?(x)$ at level k , piecewise-affine on each Farey interval and matching $?(x)$ at the endpoints. The §3 scaffold's absolutely-continuous finite levels are Padé's perch. Closure here is the strongest natural reading: the level- $k \rightarrow$ level- $(k + 1)$ update of C_k is a fixed linear map M on a finite-dimensional state v_k — §2's $\text{Aff}^+(\mathbb{R})$ closure-under-composition argument moved to a matrix algebra — with $C_k = \Phi(M^k v_0)$ for a fixed linear $\Phi: V \rightarrow C[0, 1]$. The function-valued span $W := \text{span}\{C_k : k \geq 0\}$ is then a finite-dimensional subspace of $C[0, 1]$ (it is the Φ -image of the M -invariant span of v_0 ,

itself finite-dimensional).

Salem closes that door directly. $C_k \rightarrow ?(x)$ uniformly on $[0, 1]$ (the level- k Farey points become dense and $?(x)$ is continuous), and finite-dimensional subspaces of $C[0, 1]$ are closed, so $?(x) \in W$. Each C_k is absolutely continuous, and linear combinations of absolutely continuous functions are absolutely continuous, so $W \subset \text{AC}[0, 1]$. But $?(x)$ is singular continuous (Salem 1943). Padé's ghost has nothing to land on.

Surreal §4. Closure $\implies \{C_k\}$ in finite-dim AC subspace $\implies ?(x) \in \text{AC}[0, 1]$. But $?(x)$ is singular (Salem). Padé's ghost has nothing to land on.

§ 5. The wall

§1: $z^{-1/b}$ is not a polynomial on any interval of positive length. §2: Cesàro means at every order k fail to damp the mantissa rotation, and no finite composition in $\text{Aff}^+(\mathbb{R})$ produces ψ . §3: strict concavity of ε forbids any constant-per-class corrector on the cells the orientation pairing groups together. §4: no finite-dimensional absolutely continuous subspace contains the singular continuous limit of the level-by-level corrections to $?(x)$.

What these share is a mismatch between the binary mantissa subdivision and the Stern–Brocot Farey subdivision of $[0, 1]$. ε is its residual on the dyadic side and $?(x)$ is its bijection on the scaffold side; both are infinite-dimensional objects the paper has approached from five different directions. The remainder of this section catalogs the local facts about ε the paper has been leaning on: spectral non-termination on the binade circle, transcendence at every interior dyadic, and the Baker clearance bound from rounding boundaries.

The Fourier coefficients of ε on the binade circle decay as $O(1/n^2)$ but never terminate. They never terminate because the periodic extension of ε is continuous but not C^1 at the glued endpoints: $\varepsilon'(0^+) = 1/\ln 2 - 1 \approx 0.443$, while $\varepsilon'(1^-) = 1/(2\ln 2) - 1 \approx -0.279$. A finite trigonometric polynomial is C^∞ . The derivative jump forces the series to be infinite. The $O(1/n^2)$ decay is the standard consequence of a function that is continuous but not C^1 .

A stronger fact holds at every interior point: ε is transcendental there. At every interior machine number $m = k/2^p$, the residual $\varepsilon(k/2^p) = \log_2(1 + k/2^p) - k/2^p$ is transcendental. If $\beta = \log_2(1 + m)$ were algebraic and irrational, then $2^\beta = 1 + m$ algebraic contradicts Gelfond–Schneider. If β were rational, say a/b , then $(1 + m)^b = 2^a$, so $(2^p + k)^b = 2^{a+pb}$, which would force $2^p + k$ to be a power of 2 — impossible, since $2^p < 2^p + k < 2^{p+1}$ for $0 < k < 2^p$. So β is transcendental, and $\varepsilon = \beta - m$ is transcendental. The correction is never exact.

How close does ε come to zero? Baker: $|c_1 \log \alpha_1 + c_2 \log \alpha_2| \geq \exp\{-C(\log B)^2\}$ for algebraic α_i and integer c_i with $|c_i| \leq B$. At bit precision p the linear-form coefficients have size $O(2^p)$, so $(\log B)^2 = O(p^2)$, and the clearance from any rounding boundary is at least $2^{-O(p^2)}$. The guard bit budget grows quadratically with precision and depends on no information from the previous precision.

Surreal §5. Let $m = k/2^p$, $0 < k < 2^p$. If $\varepsilon(m)$ were dyadic, $\log_2(1 + m)$ would be rational a/b , so $(2^p + k)^b = 2^{a+pb}$ — forcing $2^p + k$ to be a power of 2, impossible between consecutive ones. So $\varepsilon(m)$ is real, not dyadic. Finite-birthday surreals are dyadic. $\therefore \text{birthday}(\varepsilon(m)) = \omega$. $\square$

§ 6. The fire

Here we are. Desperate for any help. We will believe anything, give anything, pay any cost to resolve this. In our moment of despair, we are met by a demon with an offer. It knows of a game we only need to win once, somewhere in the universe. One last game, dyadic on the binary tree.

Martin's theorem tells us all Borel games are determined. The result is proved by showing that one player has a winning strategy against every possible strategy of the other.

Player II is the singular structure of $\mu?$, the same obstruction §4 just closed against Padé. If we face it alone, we already know we have lost. But this demon is special, it says. Not Descartes' demon — that one asks whether you can trust what you know, and the answer is philosophical. Not Maxwell's demon, though this also sits at a boundary between two regimes. Shannon's demon. It observes one tile at a time. It sorts.

The demon wins if it can do two things, chained. First: establish an information asymmetry across the boundary between the absolutely continuous class and the singular class. Second: aggregate what it learns. Without asymmetry there is nothing to exploit. With asymmetry but without aggregation, the demon knows something it cannot use.

Phase 1. The demon generates a smooth hallucination $\tilde{\varepsilon}$: a function on $[0,1]$ with the right general shape and the right first few Fourier modes. The hallucination is absolutely continuous. It lives on the smooth continent. Its generation cost is $O(1)$ — independent of precision, independent of the depth of the tree.

Phase 2. The demon feeds $\tilde{\varepsilon}$ into a process that interacts with ε . Not a comparison — the demon does not ask “how far is $\tilde{\varepsilon}$ from ε ?” That question costs as much as computing ε . Instead, the process performs its own work: rounding, correction, evaluation — whatever engages with $\log_2$ at the points where it must. The cost of the response is borne by the process, not by the demon.

Phase 3. The demon reads the response. Not the full computation — a partial, noisy, stochastic reading. On which side of a rounding boundary did the evaluation fall? Which narrow passage was threaded, and in which direction? One bit, a few bits, recovered at cost $O(n)$. The demon is eavesdropping, not recomputing.

The first condition is met. ε is transcendental at every interior machine number. The clearance is never zero. Information about the gap exists at every precision (Baker). The second condition is not met. Whatever the demon recovers, it must aggregate. Local observations — per-binade, per-tile — must compose into a global correction.

Bowen (2002, §2.1, eq. 2.5) gives the natural tiling-space lift when an invariant measure exists: local density ratios are integrated against that measure. But the floor does not stabilize that way on the binary tiling. These tilings point somewhere, and Bowen (2002, §2.3.1) proves there is therefore no $\text{PSL}(2, \mathbb{R})$ -invariant probability measure on their tiling space. The demon's three phases are unconstrained. Its strategy is not defeated by a resource bound, a complexity barrier, or a clever counterargument. It is defeated because the tiling's natural route to aggregation is closed.

The gap between what the representation gives and what the operations require is ε . It is transcendental at every point where it is nonzero. It lives on a tiling with no invariant probability measure. It is the reason addition is hard in scientific notation, why things must be *just like that*.

§ 7. Coda

The logarithm is free. Its residual is not. The residual is the price of using an additive grid to index a multiplicative world, and that price is controlled by the Diophantine properties of e and $\ln 2$ — two numbers that, if Schanuel's conjecture holds, are algebraically independent: no nontrivial polynomial with integer coefficients vanishes at $(e, \ln 2)$, at any scale, at any precision, down to the foam.

To get there we had to crawl a firewall we may not cross, even in our imaginations. I hope you know that now. I hope the voyage changes who you are. **That** is the beautiful inquisitiveness of the Sciences.

Surreal §7. Project the tree onto a tape. Descend left: write $-$, advance. Descend right: write $+$, advance. Ascend: retract. The walker is a Turing machine. The sign expansion is the tape.

One door remains open: whether a bounded-state walker can compute the correction at all depths without ever closing into a finite law. That door stays open because closing it in full generality would amount to deciding what bounded machines can compute. We choose a base. We compute to a precision. The machine cannot stop for us by simply closing.

Gosper (1972) proposes a machine which represents the pending arithmetic of two continued fractions as eight integer variables and a rule: consume an input term or emit an output term, each a linear map on the state. The transducer walks the Stern–Brocot tree — our tree — consuming and emitting terms one at a time. Its state grows without bound on non-Hurwitz inputs: the treewidth program's nightmare. It reaches periodicity only when the continued fraction is that of a quadratic irrational: §2's recurrence obstruction. Its operations are Möbius transformations that cannot produce the logarithmic coordinate change: §2's affine obstruction. Gosper's machine puts in the right figures and gets the right answers, term by term, for as long as you let it run. It survives by refusing finite closure.

One thread is left hanging.

§3 proves the local case: no k -local corona-invariant corrector — one whose value depends only on the cell's k -corona congruence class — equals ε at the canonical mantissa of every depth- d cell once $d \geq k + 1$. The proof rests on Dolbilin–Frettlöh's orientation pairing and the strict concavity of ε . What remains open is the streaming case: a width- q branching program reading the dyadic address bit by bit can in principle use information the corona model cannot. Whether such a machine, at fixed width q , must eventually alias on this correction function as the depth grows, is not proved here. We cite Babai, Hajnal, Szemerédi, and Turán for the existence of exponential lower bounds in the read-once branching program model. Treewidth is the natural candidate parameter: Kisfaludi-Bak et al. (2023, Proposition 1.3, restating Chepoi–Dragan–Estellon–Habib–Vaxès (2008) and Dieng–Gavoille (2009)) bound the treewidth of any n -vertex planar δ -hyperbolic graph above by $O(\delta \log n)$, and the binary tiling is planar and constant-hyperbolic. We do not prove that the correction function falls in the class their techniques reach.

We conjecture the stronger bound exists. We do not show it. You are the reader this paper was written for — this is where you can improve on what we have done. Don't let anyone tell you what's easy or hard.

§ 8. Acknowledgements

This is for my wife, my mother, and my daughter. This is because of them and everyone else who ever believed in me.

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